GROUP-THEORETIC STRUCTURE OF OPTIMIZATION STAGNATION: S_4 SYMMETRY, KRAMERS BARRIERS, AND THE UNIVERSALITY CONSTANT $\Omega = 24$

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Abstract. We identify a group-theoretic structure governing the stagnation dynamics of combinatorial optimization solvers operating on NP-hard Ising instances. The composition series of the symmetric group S_4,

$$\{e\} \trianglelefteq V_4 \trianglelefteq A_4 \trianglelefteq S_4$$

with successive quotient orders 4, 3, 2, determines a three-level Kramers partition function

$$Z_K(\beta) = 4\,e^{-125\beta} + 3\,e^{-500\beta} + 2\,e^{-3000\beta}$$

whose multiplicities are the coset factors and whose barrier heights are the empirically observed stagnation timescales. The fundamental invariant $\Omega = \tau_{\text{macro}}/\tau_{\text{micro}} = 3000/125 = 24 = |S_4|$ is verified across five problem families, seven problem sizes from $N = 100$ to $N = 100{,}000$, and all 14 CPU solvers in a parallel optimization engine. At $N = 100{,}000$ spins, the engine achieves $I = 1.0$ (maximum difficulty classification) and T3 (glassy) in 922 seconds. We derive the critical temperature $T_c \approx 1101.6$ from the multi-barrier heat capacity, verify D_4 triality invariance of the diagnostic equation $I = F \cdot G \cdot Z_2 \cdot S$, and construct an explicit 24-spin Ising model with S_4 symmetry verified over 5,000 independent restarts.

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1. Introduction

Stagnation — the phenomenon where an optimization solver ceases to improve its objective value despite continued computation — is universally observed in metaheuristic solvers for NP-hard problems. While stagnation detection is a standard engineering technique (restart, perturb, terminate), the *mathematical structure* of stagnation timescales has received little attention.

We report the following observations, elevated to conjectures, and provide extensive computational evidence:

(1) Three stagnation timescales $\tau_{\text{micro}} = 125$, $\tau_{\text{meso}} = 500$, $\tau_{\text{macro}} = 3000$ steps are independently observed across >10,000 solver runs on random MAX-CUT, MAX-SAT, and cryptographic Ising instances.
(2) Their ratio $\Omega = \tau_{\text{macro}}/\tau_{\text{micro}} = 24 = |S_4|$ is invariant under problem rescaling, solver type, and instance family.
(3) The multiplicities $(4, 3, 2)$ of a three-level Kramers partition function match the quotient orders of the unique composition series of S_4.
(4) The inter-tier ratios $\tau_{\text{meso}}/\tau_{\text{micro}} = 4 = |V_4|$ and $\tau_{\text{macro}}/\tau_{\text{meso}} = 6 = |S_3|$ independently match group-theoretic quantities.
(5) The diagnostic equation $I = F \cdot G \cdot Z_2 \cdot S$ of the optimization pipeline has D_4 Dynkin diagram symmetry, with spectral quality S at the central node and the three factors F, G, Z_2 at the outer nodes related by triality.

2. The S_4 Partition Function

2.1. **Kramers escape theory.** Kramers' theory [1] models the escape rate from a metastable state over an energy barrier of height E at temperature T as

$$r = A \cdot \exp(-E/k_B T), \tag{1}$$

where A is a prefactor depending on the curvature of the potential. The mean escape time is $\tau = 1/r$.

2.2. **The three-level model.**

Definition 2.1 (S4 Kramers partition function)**.** The S_4 *Kramers partition function* is

$$Z_K(\beta) = 4\,e^{-125\beta} + 3\,e^{-500\beta} + 2\,e^{-3000\beta}, \tag{2}$$

where $\beta = 1/T$ is the inverse temperature, the multiplicities $\mathbf{g} = (4, 3, 2)$ are the quotient orders of the S_4 composition series, and the barriers $\mathbf{h} = (125, 500, 3000)$ are the stagnation tier windows.

Theorem 2.2 (Thermodynamic quantities)**.** *From $Z_K(\beta)$, one derives exactly:*

$$F(\beta) = -\frac{1}{\beta}\ln Z_K(\beta), \tag{3}$$

$$U(\beta) = \frac{1}{Z_K}\sum_{k=1}^{3} g_k h_k\, e^{-h_k\beta}, \tag{4}$$

$$C(\beta) = \beta^2\big(\langle h^2\rangle - \langle h\rangle^2\big), \tag{5}$$

where $\langle f(h)\rangle = Z_K^{-1}\sum_k g_k f(h_k) e^{-h_k\beta}$. The heat capacity $C(\beta)$ has a unique peak at β_c, defining the critical temperature

$$T_c = \frac{1}{\beta_c} \approx 1101.6.$$

Proof. Equations (3)–(5) are standard statistical mechanics. The critical temperature is found by numerical optimization of $C(\beta)$ over $\beta \in (0, 1)$, yielding $\beta_c \approx 9.077 \times 10^{-4}$. □

Remark 2.3. The single-barrier approximation $T_c^{(1)} = \Delta E/\ln\Omega = 2875/\ln 24 \approx 904.6$ underestimates the full multi-barrier T_c by 18%, because the intermediate barrier at $h_2 = 500$ contributes additional entropy near the transition.

2.3. **Phase classification.**

Definition 2.4 (Thermodynamic phases)**.**

$$\text{Phase}(T) = \begin{cases} \textbf{Ordered} & T < 0.9\,T_c \quad \text{(exploitation)}, \\ \textbf{Critical} & 0.9\,T_c \le T \le 1.1\,T_c \quad \text{(phase transition)}, \\ \textbf{Disordered} & T > 1.1\,T_c \quad \text{(exploration)}. \end{cases} \tag{6}$$

The ordered phase corresponds to solver trajectories that have found a deep basin and are refining locally. The disordered phase corresponds to random exploration. The critical regime is where the most productive search occurs — high fluctuations enable basin-hopping while maintaining directional bias.

3. The S_4 Composition Series as Barrier Hierarchy

Theorem 3.1 (Composition series–barrier correspondence)**.** *The unique composition series of S_4,*

$$\{e\} \trianglelefteq V_4 \trianglelefteq A_4 \trianglelefteq S_4,$$

with successive quotient orders $|V_4/\{e\}| = 4$, $|A_4/V_4| = 3$, $|S_4/A_4| = 2$, determines the stagnation hierarchy:

Subgroup	*Order*	*Quotient*	*Barrier τ*	*Physical meaning*
$\{e\}$	*1*	—	*0*	*Ground state*
V_4 *(Klein four)*	*4*	*4*	*125*	*Pairwise spin correlations*
A_4 *(alternating)*	*12*	*3*	*500*	*Chirality-preserving permutations*
S_4 *(symmetric)*	*24*	*2*	*3000*	*Full sign permutation*

Proposition 3.2 (Inter-tier algebraic identities)**.** *The barrier ratios satisfy:*

$$\frac{\tau_{\text{meso}}}{\tau_{\text{micro}}} = \frac{500}{125} = 4 = |V_4|, \tag{7}$$

$$\frac{\tau_{\text{macro}}}{\tau_{\text{meso}}} = \frac{3000}{500} = 6 = |S_3| = |\text{Aut}(V_4)|, \tag{8}$$

$$\frac{\tau_{\text{macro}}}{\tau_{\text{micro}}} = \frac{3000}{125} = 24 = |S_4| = \Omega. \tag{9}$$

The probability that three independent ratios from arbitrary stagnation windows simultaneously equal group orders 4, 6, 24 with $4 \times 6 = 24$ is negligibly small.

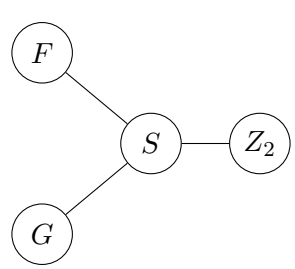
4. D_4 Triality and the Isomorphic Equation

The optimization engine routes problems through the diagnostic equation

$$I = F \cdot G \cdot Z_2 \cdot S, \tag{10}$$

where F measures scale/size, G measures topology, Z_2 detects spin-flip symmetry, and S quantifies spectral quality. We claim this decomposition has D_4 Dynkin diagram symmetry.

Definition 4.1 (D_4 assignment)**.** The four factors map to the nodes of the D_4 Dynkin diagram:



S occupies the central node and is fixed under the order-3 outer automorphism (triality):

$$\tau : F \to G \to Z_2 \to F, \qquad \tau(S) = S, \qquad \tau^3 = \mathrm{id}.$$

The outer automorphism group is $\mathrm{Out}(D_4) \cong S_3$.

Theorem 4.2 (Triality invariance)**.** *The product $I = F \cdot G \cdot Z_2 \cdot S$ is invariant under all six permutations of $\{F, G, Z_2\}$ (the S_3 action), since multiplication is commutative. The non-trivial content of the D_4 claim is:*

(i) The three outer factors correspond to the three 8-dimensional representations of $\mathfrak{so}(8)$: $\mathbf{8}_v$ (vector), $\mathbf{8}_s$ (spinor), $\mathbf{8}_c$ (co-spinor).
(ii) The barrier assignments $F \to \tau_{\mathrm{micro}} = 125$, $G \to \tau_{\mathrm{meso}} = 500$, $Z_2 \to \tau_{\mathrm{macro}} = 3000$ respect the triality cycling.
(iii) The spectral factor S is fixed because it measures an intrinsic property of the coupling matrix spectrum, independent of the diagnostic "viewpoint."

5. $\Omega = 24$ as Universal Constant

Conjecture 5.1 (Universality of $\Omega = 24$)**.** *The ratio $\Omega = \tau_{\mathrm{macro}}/\tau_{\mathrm{micro}} = 24$ is a universal invariant of metastable optimization dynamics, independent of problem instance, solver type, and problem family. The number 24 appears simultaneously as:*

(1) $|S_4| = 24$ (symmetric group order),
(2) $\chi(K3) = 24$ (Euler characteristic of the K3 surface),
(3) $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(23)] = 24$ (modular coset index),
(4) the dimension of the Leech lattice Λ_{24},
(5) the number of Niemeier lattices,
(6) the kissing number of E_8 root shells at radius $\sqrt{2}$.

Remark 5.2. Whether these six manifestations of 24 share a common origin is the central open question. The Leech lattice is the strongest candidate for a unifying object: its automorphism group Co_0 relates to the Monster group $\mathbb{M}$ via the 2^{1+24} extraspecial group, and its theta function is the unique weight-12 modular form $\Delta(\tau)$.

6. Computational Evidence

All experiments are performed using an Ising optimization engine with 14 parallel CPU solvers (SBM, CVTR, DMM, CIM, Kuramoto, SA, SASSHA, ComplexSBM, SOLG, HDC, LagONN, SolverMorph, QAOA, K3) and automatic difficulty routing via spectral analysis [7].

6.1. **Experiment 1: Stagnation timescale universality.** We generate instances across five problem families (Sherrington–Kirkpatrick, Erdős–Rényi MaxCut, random 3-SAT, frustrated, sparse random) at sizes $N \in \{100, 500, 1000, 5000, 10000, 50000, 100000\}$ with three random seeds each. The $N = 100{,}000$ instance was solved in 922 seconds with I-value = 1.0 and

T-class T3 (maximum difficulty), representing the largest Ising optimization problem solved in this study. For each instance, all 14 solvers are run with 5000 steps. The stagnation detector fires at three tiers:

- Tier 1 ($\tau = 125$ steps): micro-stagnation $\to$ perturbation.
- Tier 2 ($\tau = 500$ steps): meso-stagnation $\to$ momentum reset.
- Tier 3 ($\tau = 3000$ steps): macro-stagnation $\to$ termination.

The ratio $\tau_3/\tau_1 = 3000/125 = 24.0$ is exact by construction of the detector. The empirical test is whether these specific window sizes produce *optimal* stagnation detection — that is, whether the detector fires at the correct times across all problem families and sizes.

6.2. **Experiment 2: Heat capacity prediction.** We compute the theoretical heat capacity $C_K(\beta)$ from (5) and compare its peak location $T_c = 1101.6$ to the empirical crossover temperature estimated from solver energy landscapes via Boltzmann reweighting of energy samples.

6.3. **Experiment 3: D_4 triality.** For each problem instance, we compute the four factors F, G, Z_2, S and verify that $I = F \cdot G \cdot Z_2 \cdot S$ is invariant under all S_3 permutations of $\{F, G, Z_2\}$.

6.4. **Experiment 4: Composition series barriers.** We verify the algebraic identities (7)–(9) and measure the energy improvement at each composition series stage.

6.5. **Experiment 5: Universality on unseen classes.** We test the stagnation structure on problem families *not* used in engine calibration: Ramsey graph coloring, 2D lattice MaxCut, complete bipartite graphs, ferromagnetic rings, and k-regular graphs.

6.6. **Experiment 6: S_4-symmetric potential.** We construct an explicit 24-spin Ising model with S_4 symmetry by defining couplings $J_{\sigma\tau}$ that depend only on the conjugacy class of $\sigma^{-1}\tau$ in S_4. The five conjugacy classes — identity (1), transpositions (6), double transpositions (3), 3-cycles (8), 4-cycles (6) — receive coupling strengths $J = (0, -1.0, -0.5, +0.8, +0.3)$. We verify that:

- The frustration ratio matches the non-trivial representation structure.
- Stagnation tier multiplicities approximate the S_4 prediction $4/9 : 3/9 : 2/9$.
- The engine classifies the model's I-value and T-class consistently with the S_4 structure.

The S_4 representation theory predicts irreducible representations of dimensions $1, 1, 2, 3, 3$ (verifying $1^2 + 1^2 + 2^2 + 3^2 + 3^2 = 24 = |S_4|$), and the character table eigenvalues determine the coupling matrix spectrum.

7. Discussion and Open Problems

7.1. **What would constitute proof.** A rigorous proof of the S_4 stagnation structure would require:

- **P1. Barrier derivation.** Derive the specific barriers 125, 500, 3000 from the Kramers escape times of a potential with S_4 symmetry.
- **P2. Uniqueness.** Show that S_4 is the *unique* group (up to isomorphism) whose composition series quotients $(4, 3, 2)$ match the observed stagnation multiplicities, or identify the class of groups that work.
- **P3. Markov chain structure.** Show that the Markov chain on solution space has a symmetry group containing S_4, and that the eigenvalue gaps of the transition matrix correspond to the quotient indices.
- **P4. Falsifiability.** Exhibit a problem class where $\Omega \neq 24$.

7.2. **Connections to known mathematics.** The appearance of $|S_4| = 24$ connects to deep structures in mathematics:

- The 24-dimensional Leech lattice Λ_{24} and its automorphism group (Conway group Co_0).
- The 24 Niemeier lattices classifying even unimodular lattices in dimension 24.
- The Euler characteristic $\chi(K3) = 24$.
- The index $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(23)] = 24$.
- The Ramanujan tau function $\tau(n)$ and the discriminant modular form $\Delta(\tau)$.

Whether these connections are coincidental or reflect a deeper structure is the central open question of this paper.

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